

Homework — 1.5.1 Computing Related Legislation — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] Completed table:

Scenario	Act that applies
Nurse looks up a celebrity's medical records out of curiosity	Computer Misuse Act 1990 (unauthorised access to computer material — Offence 1)
Photographs copied and used without permission in a paid app	Copyright, Designs and Patents Act 1988
Names and emails kept for ten years "just in case"	Data Protection Act 2018 (data kept no longer than necessary)
Authorised interception of a suspect's phone calls	Regulation of Investigatory Powers Act 2000

Mark: 2 correct rows per mark (4 rows → 2 marks). Act name must be identifiable; the bracketed detail is not required.

2. [3] Completed paragraph, in order of the blanks:

1. **data subject**
2. **data controller**
3. **data processor**
4. **no longer**
5. **up to date**
6. **securely**

Unused distractors: data owner, indefinitely, publicly.

Mark: 2 correct blanks per mark (6 blanks → 3 marks).

3. [3] Completed true/false table:

Statement	True / False	Correction if false
Unauthorised access is an offence even if nothing is changed or stolen.	True	— (this is CMA Offence 1.)
A program must be registered before copyright protects it.	False	Copyright arises automatically on creation — no registration is needed.
Open source means the developer gives up their copyright.	False	The developer keeps the copyright ; the open-source licence is how they grant others permission to copy/modify the work.
The DPA 2018 applies to personal data of living individuals.	True	—
RIPA 2000 lets any police officer authorise interception.	False	Only designated senior officials may authorise interception, and only for legitimate purposes (e.g. national security, serious crime) — providing oversight and preventing abuse.
Distributing a file-deleting virus breaches the unauthorised-modification offence.	True	— (CMA Offence 3: unauthorised modification/impairment of computer material.)

Mark: 2 correct rows per mark (6 rows → 3 marks). A false row scores only if a valid correction is given.

4. [2] 1 mark: Computer Misuse Act 1990, **Offence 3** (unauthorised modification of computer material). 1 mark: deleting/altering the database is an unauthorised *modification*; the Act is about *authorisation* and the technician was no longer authorised to act on the system (and never authorised to destroy data) — it does not require a password to be stolen, only that the modification was unauthorised. (*Do not accept Offence 1 or 2.*)

5. [4] Levels-of-response "**discuss**" question — mark the **whole answer**, not point-by-point.

Indicative content — - **FOR the powers**: serious crime and terrorism are increasingly planned using digital communications, so investigators need lawful interception; the power to demand encryption keys stops evidence being locked away; restricting *who may authorise* use provides accountability and ensures the powers are proportionate and used only for legitimate purposes. - **AGAINST / concerns**: interception is a serious intrusion into privacy of people who may be innocent; risk of misuse or "mission creep" (powers used for minor matters); demanding keys can compromise everyone's data security; chilling effect on free communication. - **Justified conclusion**: a clear, reasoned position, ideally conditional — e.g. "*the powers are acceptable only while authorisation remains restricted to senior officials, is limited to serious purposes and is subject to oversight.*"

Brief level descriptors: - **Level 1 (1)**: one-sided or fragmented points; no conclusion. - **Level 2 (2–3)**: both sides touched on but limited development; weak conclusion. - **Level 3 (4)**: balanced discussion of investigative benefit **and** privacy cost, referring to the authorisation safeguard, with a justified conclusion.

Part B (6 marks)

6. [2] 1 mark + 1 mark for a clear difference, e.g. symmetric encryption uses the **same single key** to encrypt and decrypt (which must be shared securely) [1]; asymmetric uses a **key pair** — a public key to encrypt and a different private key to decrypt, so no secret key need be shared [1]. (*Accept: symmetric is faster but has the key-distribution problem; asymmetric is slower but solves key exchange.*)

7. [2] 1 mark: a stack is **LIFO** — the last item pushed on is the first popped off (removed from the top). 1 mark: a queue is **FIFO** — the first item enqueued is the first dequeued (removed from the front).

8. [2] 1 mark + 1 mark, any one developed point: high-level code is closer to English / easier and faster to write, read and maintain [1]; it is portable across different processors whereas assembly is tied to one architecture / fewer instructions needed / built-in features (data structures, libraries) handle low-level detail for you [1].

Total: 14 + 6 = 20 marks